

PrepPilot — Readable Expanded Syllabus

Every chapter/topic is intentionally wrapped into multiple lines and bullet points. No single-line topic dumps.

JEE Main 2026

Mathematics

Sets, Relations and Functions

- Sets
- Union
- Intersection
- Complement
- Power set
- Relations
- Equivalence relations
- Functions
- One-one
- Into
- Onto
- Composition

Complex Numbers and Quadratic Equations

- Complex numbers as ordered pairs
- $A+ib$
- Argand diagram
- Algebra
- Modulus
- Argument
- Quadratic equations
- Roots
- Coefficients
- Nature of roots
- Formation from roots

Matrices and Determinants

- Matrices
- Operations
- Types
- Determinants $2 \times 2 / 3 \times 3$
- Evaluation
- Triangle area
- Adjoint
- Inverse
- Consistency
- Simultaneous linear equations

Permutations and Combinations

- Fundamental counting principle
- Permutations
- Combinations
- $P(n,r)$
- $C(n,r)$
- Applications

Binomial Theorem

- Positive integral index
- General term
- Middle term
- Applications

Sequence and Series

- AP
- GP
- Insertion of AM/GM
- AM-GM relation

Limit, Continuity and Differentiability

- Real functions
- Algebra of functions
- Polynomial/rational/trigonometric/log/exponential functions
- Inverse functions
- Graphs
- Limits
- Continuity
- Differentiation
- Product/quotient/sum/difference
- Trig/inverse trig/log/exponential/composite/implicit derivatives
- Second derivatives
- Rate of change
- Monotonicity
- Maxima/minima

Integral Calculus

- Anti-derivative
- Algebraic/trigonometric/exponential/log integrals
- Substitution
- Parts
- Partial fractions
- Trig identities
- Fundamental theorem
- Definite integrals
- Area bounded by curves

Differential Equations

- Ordinary differential equations
- Order
- Degree
- Separation of variables
- Homogeneous equations
- Linear first-order equations

Coordinate Geometry

- Cartesian coordinates
- Distance
- Section
- Locus
- Slope
- Parallel/perpendicular lines
- Intercepts
- Line equations
- Intersections
- Angles
- Concurrence
- Distance point-line
- Triangle centres
- Circle
- Parabola
- Ellipse
- Hyperbola

Three Dimensional Geometry

- Coordinates
- Distance
- Section
- Direction ratios/cosines
- Angle between lines
- Line equation
- Skew lines
- Shortest distance

Vector Algebra

- Vectors/scalars
- Addition
- Components in 2D/3D
- Scalar product
- Vector product

Statistics and Probability

- Dispersion
- Mean
- Median
- Mode
- Standard deviation
- Variance
- Mean deviation
- Event probability
- Addition/multiplication
- Bayes theorem
- Random-variable distribution

Trigonometry

- Trig identities/functions
- Inverse trig functions
- Properties

Physics

Units and Measurements

- Units
- SI
- Fundamental/derived units
- Least count
- Significant figures
- Errors
- Dimensions
- Dimensional analysis

Kinematics

- Frame of reference
- Straight-line motion
- Speed/velocity
- Uniform/non-uniform
- Average speed
- Instantaneous velocity
- Acceleration
- Position-time and velocity-time graphs
- Relative velocity
- Projectile
- Circular motion

Laws of Motion

- Force/inertia
- Newton I/II/III
- Momentum
- Impulse
- Momentum conservation
- Concurrent-force equilibrium
- Static/kinetic friction
- Rolling friction
- Centripetal force
- Level/banked road

Work, Energy and Power

- Constant/variable force work
- Kinetic/potential energy
- Work-energy theorem
- Power
- Spring energy
- Mechanical-energy conservation
- Conservative/non-conservative force
- Vertical circle
- Elastic/inelastic collisions
- 1D/2D collisions

Rotational Motion

- Centre of mass
- Rotation concepts
- Moment/torque
- Angular momentum
- Conservation
- Moment of inertia
- Radius of gyration
- Parallel/perpendicular axes
- Rigid-body equilibrium/rotation
- Linear vs rotational

Gravitation

- Universal law
- G variation with altitude/depth
- Kepler
- Gravitational potential/energy
- Escape velocity
- Satellites
- Orbital velocity
- Period
- Energy

Properties of Solids and Liquids

- Elasticity
- Stress/strain
- Hooke law
- Young/bulk/rigidity moduli
- Pressure
- Pascal law
- Viscosity
- Stokes
- Terminal velocity
- Streamline/turbulent
- Bernoulli
- Surface tension
- Capillary rise
- Heat
- Expansion
- Specific heat
- Calorimetry
- Phase change
- Latent heat
- Conduction
- Convection
- Radiation

Thermodynamics

- Thermal equilibrium
- Temperature
- Zeroth law
- Heat
- Work
- Internal energy
- First law
- Isothermal
- Adiabatic
- Second law
- Reversible/irreversible

Kinetic Theory of Gases

- Perfect-gas equation
- Compression work
- Assumptions
- Pressure
- Kinetic temperature
- RMS speed
- Degrees of freedom
- Equipartition
- Specific heats
- Mean free path
- Avogadro number

Oscillations and Waves

- Periodic motion
- Time period
- Frequency
- SHM
- Phase
- Spring
- SHM energy
- Pendulum
- Waves
- Longitudinal/transverse
- Progressive waves
- Superposition
- Reflection
- Standing waves
- Strings
- Organ pipes
- Harmonics
- Beats

Electrostatics

- Charge/conservation
- Coulomb
- Multiple charges
- Superposition
- Continuous distribution
- Field
- Field lines
- Dipole
- Torque
- Flux
- Gauss law
- Line/plane/spherical shell
- Potential
- Potential difference
- Equipotential
- Potential energy
- Conductors
- Dielectrics
- Polarisation
- Capacitance
- Capacitor combinations
- Parallel plate
- Dielectric
- Energy

Current Electricity

- Drift velocity
- Mobility
- Ohm law
- Resistance
- I-V
- Energy/power
- Resistivity/conductivity
- Resistor combinations
- Temperature dependence
- Internal resistance
- Potential/EMF
- Cells
- Kirchhoff
- Wheatstone
- Metre bridge

Magnetic Effects of Current and Magnetism

- Biot-Savart
- Circular loop
- Ampere
- Straight wire
- Solenoid
- Lorentz force
- Current conductor
- Parallel currents
- Ampere definition
- Torque
- Galvanometer
- Sensitivity
- Ammeter/voltmeter
- Magnetic dipole
- Bar magnet
- Axis/perpendicular-axis fields
- Magnetic materials
- Temperature

Electromagnetic Induction and Alternating Currents

- Faraday
- Induced EMF/current
- Lenz
- Eddy currents
- Self/mutual inductance
- AC peak/RMS
- Reactance
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- LCR
- Resonance
- Power
- Wattless current
- Generator
- Transformer

Electromagnetic Waves

- Displacement current
- Properties
- Transverse nature
- Spectrum from radio to gamma
- Applications

Optics

- Reflection
- Spherical mirrors
- Mirror formula
- Refraction
- Thin lens
- Lens maker
- TIR
- Magnification
- Power
- Lens combinations
- Prism
- Microscope
- Telescope
- Wavefront
- Huygens
- Interference
- Young double-slit
- Fringe width
- Coherence
- Diffraction
- Central maximum
- Polarisation
- Brewster
- Polaroids

Dual Nature of Matter and Radiation

- Dual nature
- Photoelectric effect
- Hertz/Lenard observations
- Einstein equation
- Particle nature
- Matter waves
- De Broglie

Atoms and Nuclei

- Alpha scattering
- Rutherford
- Bohr
- Energy levels
- Hydrogen spectrum
- Nuclear composition/size
- Atomic masses
- Mass-energy
- Mass defect
- Binding energy
- Fission
- Fusion

Electronic Devices

- Semiconductors
- Diode I-V
- Rectifier
- LED
- Photodiode
- Solar cell
- Zener
- Voltage regulator
- OR/AND/NOT/NAND/NOR

Experimental Skills

- Vernier
- Screw gauge
- Pendulum
- Metre scale moments
- Young modulus
- Surface tension
- Viscosity
- Sound resonance tube
- Specific heat
- Metre bridge
- Ohm law
- Galvanometer
- Focal length
- Prism deviation
- Refractive index
- P-n junction
- Zener
- Component identification

Chemistry

Some Basic Concepts in Chemistry

- Matter
- Dalton
- Atom/molecule/element/compound
- Laws of combination
- Atomic/molecular mass
- Mole
- Molar mass
- Percentage composition
- Empirical/molecular formula
- Equations
- Stoichiometry

Atomic Structure

- EM radiation
- Photoelectric effect
- H spectrum
- Bohr
- Energy/radius
- Limits
- Dual nature
- De Broglie
- Uncertainty
- Quantum model
- Orbitals
- Quantum numbers
- Orbital shapes
- Spin
- Aufbau
- Pauli
- Hund
- Configuration

Chemical Bonding

- Kossel-Lewis
- Ionic bonding
- Lattice enthalpy
- Covalent bonding
- Electronegativity
- Fajan
- Dipole
- VSEPR
- Shapes
- VBT
- Hybridisation
- Resonance
- MO theory
- LCAO
- Sigma/pi
- Bond order/length/energy
- Metallic bonding
- H-bond

Chemical Thermodynamics

- System/surroundings
- Extensive/intensive
- State functions
- Entropy
- First law
- Work
- Heat
- Internal energy
- Enthalpy
- Heat capacities
- Hess law
- Bond dissociation
- Combustion
- Formation
- Atomisation
- Sublimation
- Phase transition
- Hydration
- Ionisation
- Solution enthalpy
- Second law
- Spontaneity
- Gibbs

Solutions

- Molality
- Molarity
- Mole fraction
- Percentage
- Vapour pressure
- Raoult
- Ideal/non-ideal
- Colligative properties
- Molar mass
- Abnormal values
- Van't Hoff factor

Equilibrium

- Physical equilibrium
- Henry law
- Chemical equilibrium
- Mass action
- K_p/K_c
- Gibbs relation
- Factors
- Le Chatelier
- Weak/strong electrolytes
- Acid/base theories
- Ionisation
- PH
- Common ion
- Hydrolysis
- K_{sp}
- Buffers

Redox Reactions and Electrochemistry

- Oxidation/reduction
- Oxidation number
- Balancing
- Conduction
- Molar conductivity
- Kohlrausch
- Electrolytic/galvanic cells
- Electrode potentials
- EMF
- Nernst
- Gibbs relation
- Dry cell
- Lead accumulator
- Fuel cell

Chemical Kinetics

- Rate
- Concentration/temperature/pressure/catalyst effects
- Elementary/complex
- Order
- Molecularity
- Rate law
- Zero/first order
- Half-life
- Arrhenius
- Activation energy
- Collision theory

Classification and Periodicity

- Modern periodic law
- Table
- Electronic configuration
- Atomic/ionic radius
- Ionisation enthalpy
- Electron-gain enthalpy
- Valence
- Oxidation state
- Electronegativity
- Reactivity

Hydrogen

- Position
- Occurrence
- Isotopes
- Preparation
- Properties
- Uses
- Hydrides
- Water
- Heavy water
- Hydrogen peroxide
- Fuel

s-Block Elements

- Groups 1/2
- Trends
- Compounds
- Reactions

p-Block Elements

- Groups 13-18
- Configuration
- Trends
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d- and f-Block Elements

- Transition elements
- Oxidation states
- Complexes
- Catalysis
- Magnetic behaviour
- Lanthanoids
- Actinoids
- Applications

Coordination Compounds

- Werner
- Terminology
- Nomenclature
- Isomerism
- Bonding
- Stability
- Applications

Organic Chemistry Principles

- IUPAC
- Structural/stereo isomerism
- Purification
- Analysis
- Electronic effects
- Intermediates
- Reaction types

Hydrocarbons

- Alkanes
- Alkenes
- Alkynes
- Aromatic hydrocarbons
- Preparation
- Reactions

Organic Compounds Containing Halogens

- Haloalkanes
- Haloarenes
- C-X bond
- Preparation
- Substitution
- Elimination
- Polyhalogen compounds

Alcohols Phenols and Ethers

- Classification
- Nomenclature
- Preparation
- Physical properties
- Reactions
- Phenol acidity
- Ethers
- Uses

Aldehydes Ketones and Carboxylic Acids

- Nomenclature
- Carbonyl structure
- Preparation
- Physical properties
- Reactions
- Carboxylic acid structure
- Preparation
- Properties
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- Uses

Amines

- Structure
- Classification
- Nomenclature
- Preparation
- Properties
- Reactions
- Diazonium salts
- Synthesis use

Biomolecules and Polymers

- Carbohydrates
- Proteins
- Enzymes
- Vitamins
- Nucleic acids
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Atomic Structure

- EM radiation
- Photoelectric effect
- H spectrum
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- Energy/radius
- Limits
- Dual nature
- De Broglie
- Uncertainty
- Quantum model
- Orbitals
- Quantum numbers
- Orbital shapes
- Spin
- Aufbau
- Pauli
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Chemical Bonding

- Kossel-Lewis
- Ionic bonding
- Lattice enthalpy
- Covalent bonding
- Electronegativity
- Fajan
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Amines

- Structure
- Classification
- Nomenclature
- Preparation
- Properties
- Reactions
- Diazonium salts
- Synthesis use

Biomolecules and Polymers

- Carbohydrates
- Proteins
- Enzymes
- Vitamins
- Nucleic acids
- Polymers

Biology

Diversity in Living World

- Living world
- Taxonomy
- Biological classification
- Plant kingdom
- Animal kingdom

Structural Organisation

- Morphology and anatomy of flowering plants
- Tissues
- Animal organisation

Cell Structure and Function

- Cell
- Organelles
- Biomolecules
- Membrane
- Cell cycle
- Mitosis
- Meiosis

Plant Physiology

- Transport
- Mineral nutrition
- Photosynthesis
- Respiration
- Plant growth and development

Human Physiology

- Breathing
- Circulation
- Excretion
- Locomotion
- Neural control
- Chemical coordination

Reproduction

- Flowering-plant reproduction
- Human reproduction
- Reproductive health

Genetics and Evolution

- Inheritance
- Molecular basis
- DNA/RNA
- Replication
- Transcription
- Translation
- Evolution

Biology and Human Welfare

- Health/disease
- Microbes in human welfare
- Immunity
- Human welfare applications

Biotechnology

- Recombinant DNA
- Genetic engineering
- Health/agriculture applications
- Biosafety

Ecology and Environment

- Organisms/environment
- Population interactions
- Ecosystem
- Biodiversity
- Conservation
- Environmental issues

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Mathematics

Sets, Relations and Functions

- Sets
- Relations
- Functions
- Composition

Algebra

- Complex numbers
- Quadratic equations
- Sequences/series
- Binomial
- Permutations/combinations
- Matrices/determinants

Trigonometry

- Identities
- Equations
- Inverse functions

Coordinate Geometry

- Straight lines
- Circles
- Conics
- 3D

Differential Calculus

- Limits
- Continuity
- Differentiability
- Applications

Integral Calculus

- Indefinite/definite integration
- Area
- Differential equations

Probability and Statistics

- Probability
- Conditional probability
- Bayes
- Statistics

Vectors

- Vector operations
- Applications

Physics

General Physics

- Units
- Dimensions
- Errors
- Significant figures
- Vectors

Mechanics

- Kinematics
- Newton laws
- Friction
- Work-energy
- Momentum
- Centre of mass
- Rotation
- Gravitation
- Fluids

Thermal Physics

- Thermal expansion
- Calorimetry
- Heat transfer
- Thermodynamics
- Kinetic theory

Waves and Oscillations

- SHM
- Waves
- Sound
- Standing waves
- Beats

Electromagnetism

- Electrostatics
- Current electricity
- Magnetism
- EMI
- AC

Optics

- Geometrical optics
- Interference
- Diffraction
- Polarisation

Modern Physics

- Photoelectric effect
- Atoms
- Nuclei
- Radioactivity
- Semiconductors

Chemistry

General Topics

- Atoms/molecules
- Dalton theory
- Mole concept
- Formulae
- Balanced equations
- Stoichiometry
- Mole fraction
- Molarity
- Molality
- Normality

States of Matter: Gases and Liquids

- Gas laws
- Ideal gas
- Van der Waals
- Kinetic theory
- Average/RMS/most probable speeds
- Partial pressures
- Diffusion
- Intermolecular interactions
- Vapour pressure
- Surface tension
- Viscosity

Atomic Structure

- Bohr
- H spectrum
- De Broglie
- Uncertainty
- Quantum numbers
- Wave function
- Probability density
- Orbital shapes
- Aufbau
- Pauli
- Hund

Chemical Bonding

- Orbital overlap
- Covalent bond
- Hybridisation
- MO diagrams up to Ne₂
- Hydrogen bond
- Polarity/dipole moment
- VSEPR

Thermodynamics

- Intensive/extensive
- State functions
- First/second laws
- Internal energy
- PV work
- Heat
- Enthalpy
- Hess law
- Entropy
- Gibbs
- Equilibrium/spontaneity

Chemical and Ionic Equilibrium

- Mass action
- K_p/K_c
- Reaction quotient
- Le Chatelier
- K_{sp}
- Common ion
- P_H
- Buffers
- Brønsted/Lewis acids/bases
- Hydrolysis

Electrochemistry

- Cells
- Standard potentials
- Nernst
- Electrochemical series
- Faraday laws
- Conductance
- Kohlrausch
- Batteries
- Fuel cells
- Corrosion

Chemical Kinetics

- Rates
- Order
- Molecularity
- Rate law
- Half-life
- Zero/first order
- Arrhenius
- Activation energy
- Catalysis

Solid State

- Crystalline state
- Seven systems
- Unit-cell parameters
- Close packing
- FCC/BCC/HCP
- Defects

Solutions

- Henry
- Raoult
- Ideal solutions
- Colligative properties
- Van't Hoff factor

Surface Chemistry

- Physisorption
- Chemisorption
- Freundlich
- Colloids
- Emulsions
- Surfactants
- Micelles

Periodicity and Blocks

- Periodic law
- Configuration
- Radius
- Ionisation enthalpy
- Electron-gain enthalpy
- Valence
- Oxidation states
- Electronegativity
- s/p/d/f blocks

Hydrogen

- Position
- Isotopes
- Preparation
- Properties
- Hydrides
- Water
- Hydrogen peroxide
- Fuel

Coordination Chemistry

- Ligands
- Coordination number
- Nomenclature
- Isomerism
- Bonding
- Stability
- Applications

Organic Chemistry

- Nomenclature
- Isomerism
- Mechanisms
- Hydrocarbons
- Halogen compounds
- Alcohols/phenols/ethers
- Carbonyls
- Carboxylic acids
- Amines
- Biomolecules/polymers